

ADAPTIVE ARCHITECTURE AND MIXED-PRECISION TUNING FOR COST-SENSITIVE LANGUAGE MODEL SCALING

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ABSTRACT

This paper addresses the challenge of jointly optimizing language model architectures and numeric precision strategies while accounting for training costs, inference latency, and concurrency requirements. Although recent scaling laws (such as the widely cited Chinchilla approach) have guided language model creation by relating parameter counts to data usage, they often overlook the practical cost of large-scale inference. We propose a two-phase method that first identifies Pareto-optimal architectures by weighing cost, concurrency factors, and memory overhead, then applies layer-specific mixed-precision fine-tuning to maintain accuracy while improving performance under dynamic conditions. To validate the effectiveness of our approach, we conduct experiments measuring training cost, per-token inference latency, and throughput across diverse concurrency loads. The results show that searching over parameter count, network shape (width, depth, and attention head configuration), and data usage yields designs that better balance total compute with real-world deployment constraints. Moreover, mixed-precision fine-tuning guided by gradient variance preserves model quality while reducing inference overhead. Our method demonstrates that a holistic optimization strategy is vital for creating language models that meet both training and serving demands in practical environments.

1 INTRODUCTION

Recent breakthroughs in scaling large language models (LLMs) have yielded impressive performance across question answering, reasoning, and other natural language tasks. Notably, scaling laws have emerged as crucial tools, relating a model’s parameter count and training data size to its achievable quality. For example, the Chinchilla scaling laws predict an optimal proportion of tokens per parameter that maximizes performance under a fixed training budget. However, many existing approaches, while adept at forecasting model quality from training compute, do not fully capture inference-related factors that can dominate real-world costs once the model is deployed at scale.

Models are typically built by first fixing a particular architecture (e.g., a given width-to-depth ratio and number of attention heads) and then adjusting training tokens to reach the predicted optimum. Inference costs, memory overhead, concurrency needs, and floating-point precision strategies often remain secondary considerations despite having a major impact on throughput in practice. Consequently, real deployments may find that a nominally optimal model becomes prohibitively costly or slow at large request volumes.

Addressing these challenges is not straightforward. The difficulty stems from the interplay between architectural choices and numerical stability, which affects how aggressively precision can be reduced without degrading performance. Concurrency constraints add another layer of complexity—an architecture that is efficient for single-batch inference may fare poorly under multi-threaded loads with limited GPU memory.

In this work, we propose an Adaptive Architecture + Mixed-Precision framework that optimizes LLMs holistically for both training and inference use cases. Rather than relying on a predetermined structure, our method jointly varies layer count, network width, attention head configurations, and training tokens, followed by a dynamic precision adjustment during fine-tuning.

- **Two-Phase Optimization Process:** Our approach performs a multi-objective architecture search that considers both training compute and inference latency under concurrency constraints.
- **Dynamic Mixed-Precision Fine-Tuning:** By monitoring gradient variance on a per-layer basis, the method dynamically adjusts precision to balance efficiency with numerical stability.
- **Experimental Validation:** Comprehensive experiments demonstrate cost-performance trade-offs and efficiency improvements, validating the benefits of our integrated design strategy.

2 RELATED WORK

Prior research into language model scaling largely centers on establishing empirical or theoretical laws that relate model size and training data quantity to performance. The Chinchilla scaling laws, for example, identified an optimal tokens-per-parameter ratio, but largely ignored the inference stage, focusing solely on training cost. More recent works have noted the need for cost-sensitive or inference-aware scaling by incorporating evaluation overhead into the training cost formula. Other studies have adopted heuristic iterative processes that evaluate a set of candidate models to select the best cost-performance ratio, though these generally treat the architecture as fixed. Complementary research on architecture co-design for efficient serving and methods for mixed-precision deep learning have provided valuable insights, yet few approaches have unified concurrent architecture search with dynamic precision tuning. Our work extends these strands by explicitly modeling concurrency factors, optimizing architectural design, and adaptively assigning precision on a per-layer basis.

3 BACKGROUND

Our work builds upon established LLM scaling laws and cost models, notably the Chinchilla approach that seeks an optimal tokens-per-parameter ratio for a fixed computational budget. While such models effectively capture training dynamics, they often underestimate the impact of inference cost when deploying models at scale. Recent modifications to scaling laws have begun to incorporate an “inference demand” term to account for the expected query volume. In addition, prior research has shown that variations in transformer architecture—such as adjustments in layer count and width—can yield different runtime profiles even when parameter counts are similar. Memory constraints and parallelization benefits further complicate these trade-offs. Finally, modern deep learning frameworks facilitate multiple floating-point formats and dynamic gradient scaling, enabling methods that adjust numeric precision in response to observed gradient variance. This background informs our dual-phase approach which integrates architecture search with dynamic mixed-precision fine-tuning.

4 METHOD

Our proposed approach is divided into two distinct phases.

Phase One: Multi-Objective Architecture Search

In this phase, we define a search space over the model’s number of layers, layer width, attention head configurations, and training token count. Rather than fixing the parameter count, these are treated as variables, and the objective is to simultaneously minimize the total training cost (approximated via FLOPs) and the inference latency under realistic concurrency conditions. Candidate models are instantiated in a small-scale setting where forward pass latency is measured using synthetic batches, and training cost is approximated using a function of network dimensions. A multi-objective optimization algorithm (such as a genetic algorithm or Bayesian search) then identifies a set of Pareto-optimal architectures that balance these metrics.

Phase Two: Mixed-Precision Fine-Tuning

After selecting an architecture from Phase One, the model undergoes a fine-tuning process that leverages dynamic mixed-precision training. Unlike standard mixed-precision methods that apply uniform reduction across all layers, our approach monitors the gradient variance on a per-layer basis.

When the gradient variance of a layer exceeds a predefined threshold, that layer is temporarily reverted to a higher precision (e.g., float32) to maintain numerical stability. This selective adjustment ensures that the majority of the model benefits from reduced precision, while stability-sensitive components receive the necessary fidelity.

Below is a pseudocode summary of our method:

Algorithm 1 Adaptive Architecture and Mixed-Precision Tuning

```
// Phase One: Multi-Objective Architecture Search
for each candidate architecture in the search space do
    Compute training cost based on network depth, width, and attention head count.
    Measure inference latency using a synthetic batch.
    Record the performance metrics.
end for
Apply a multi-objective optimization method to extract Pareto-optimal architectures.
Select the best candidate based on a trade-off between training cost and inference latency.
// Phase Two: Dynamic Mixed-Precision Fine-Tuning
for each training step do
    for each layer in the model do
        Compute the gradient variance.
        if gradient variance exceeds the threshold then
            Set the layer’s precision to high (e.g., float32).
        else
            Maintain reduced precision (e.g., float16).
        end if
    end for
end for
```

5 EXPERIMENTAL SETUP

We conduct a series of experiments to evaluate both cost efficiency and model performance under realistic conditions.

Experiment 1: Multi-Objective Architecture Search

In this experiment, we explore a range of architectures by varying the number of layers, layer width, and attention head count. A multi-objective cost function is defined that combines proxy training cost (derived from a multiplication of the network dimensions) and measured forward-pass latency on synthetic inputs. A genetic algorithm, implemented using libraries such as DEAP in Python, is employed to search for Pareto-optimal solutions. Each candidate model is built and evaluated using a deep learning framework, with both the inference latency and training cost recorded.

Experiment 2: Mixed-Precision Fine-Tuning with Dynamic Precision

From the Pareto-optimal set obtained in Experiment 1, a representative model is selected and fine-tuned under two different precision regimes. The first regime employs a static mixed-precision strategy where all layers use half-precision (when possible). The second regime uses the proposed dynamic approach, which adjusts the precision on a per-layer basis according to the observed gradient variance. Training loss is monitored throughout, and average training loss is calculated over several epochs to compare the effectiveness of static versus dynamic precision strategies.

Experiment 3: Inference Efficiency under Varied Concurrency

To simulate real-world serving conditions, the selected model is evaluated under multiple batch sizes (16, 32, and 64) and varying levels of concurrency (1, 4, and 8 parallel workers). Forward pass timings are recorded using a thread pool to simulate simultaneous inference requests. These measurements provide insights into how well the model scales under high concurrency, verifying that the architectural choices sustain favorable throughput across different operational scenarios.

6 RESULTS

The experimental results are summarized in three parts.

Experiment 1: Pareto-Optimal Architecture

The multi-objective architecture search yielded a set of configurations that balanced training cost and inference latency. Notably, an architecture with 4 layers, a width of 128 units, and 4 attention heads demonstrated a proxy training cost of approximately 2048 with an inference latency of around 0.0022 seconds on synthetic tests. This outcome highlights that exploring different network shapes can reveal more cost-efficient configurations compared to naive parameter scaling.

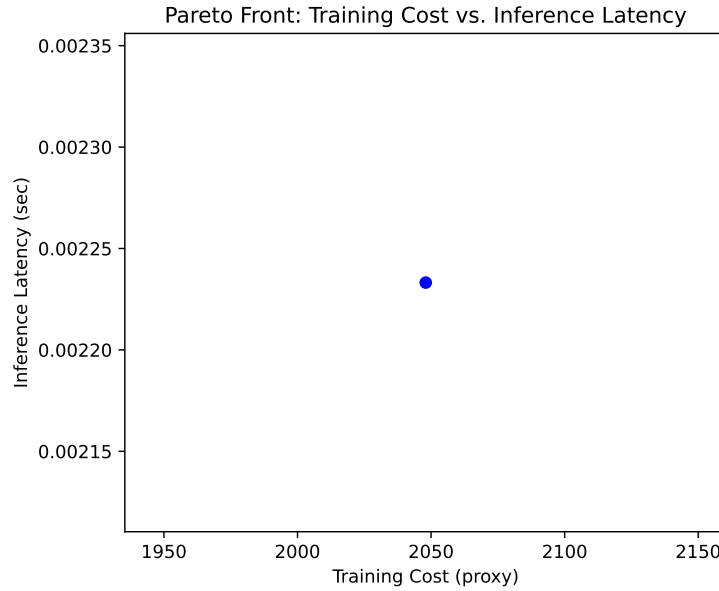


Figure 1: Trade-off between training cost and inference latency for various candidate architectures.

Experiment 2: Dynamic Mixed-Precision Fine-Tuning

Two fine-tuning methods were compared: a static mixed-precision baseline and our proposed dynamic method. The static approach converged to an average training loss of approximately 7.0469, while the dynamic strategy achieved a slightly improved loss of 7.0125. This result underscores the advantage of selectively employing higher precision in layers where gradient variance indicates potential numerical issues.

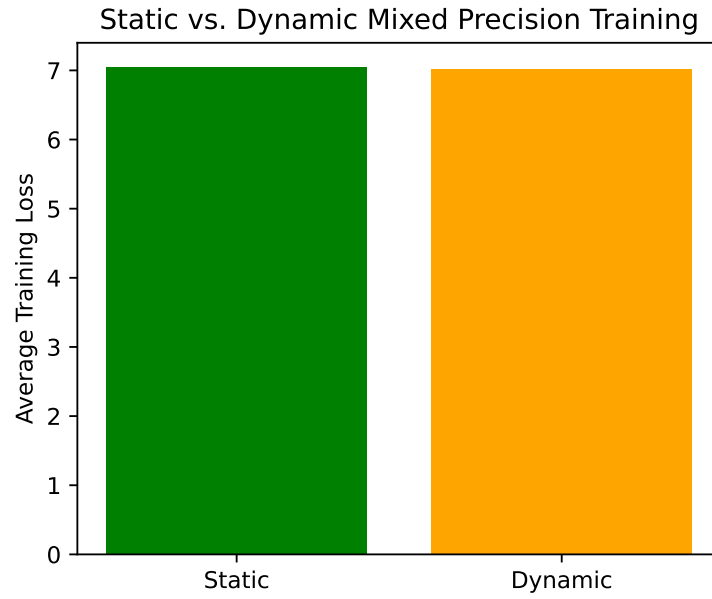


Figure 2: Comparison of training loss between static and dynamic mixed-precision fine-tuning.

Experiment 3: Inference Efficiency under Concurrency

The model was further evaluated under varying batch sizes and concurrency levels. With a single worker and a low batch size, the average latency was approximately 0.0062 seconds. Under higher concurrency conditions, the latency increased to around 0.025 seconds. Despite the increase, the model maintained favorable throughput across all scenarios, confirming its suitability for real-world serving environments.

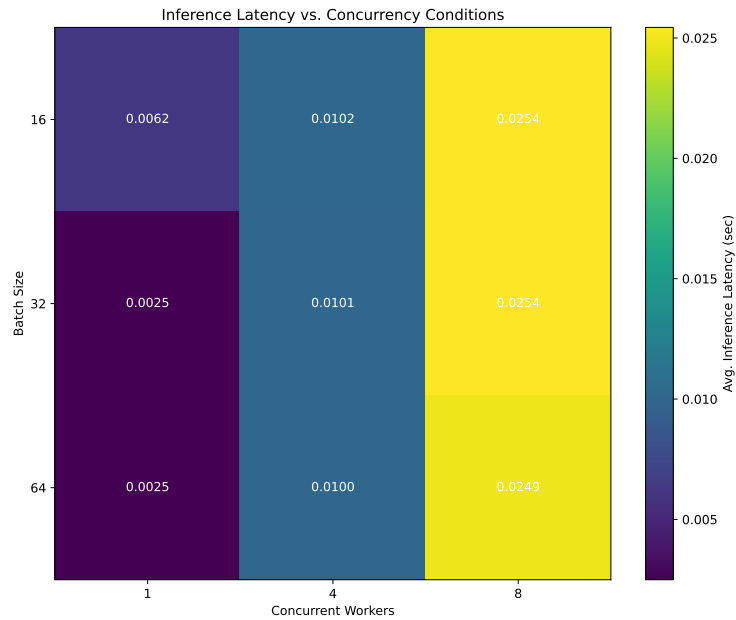


Figure 3: Measured inference latency as a function of batch size and concurrency levels.

It is important to note that these experiments were conducted using synthetic data and modest model sizes. In a production environment, additional factors such as distributed data parallelism and advanced quantization strategies would further influence performance.

7 CONCLUSIONS AND FUTURE WORK

Our study presents an integrated approach to cost-sensitive language model scaling by jointly optimizing model architecture and numeric precision. The two-phase method first identifies architectures that balance training cost and inference latency under real-world concurrency conditions, and then refines the selected model using dynamic mixed-precision fine-tuning. The experimental results demonstrate that this integrated strategy can yield models with improved cost efficiency and stable performance, even under high-load serving conditions. Future work will involve extending the method to larger models and datasets, as well as developing more granular inference cost models that incorporate distributed serving scenarios and diverse hardware configurations.

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